MILITARY SPECIFICATION

READ AMPLIFIER: 10526589

1. SCOPE

1.1 This specification covers one type of read amplifier circuit board, part of the Gun Direction Computer: M18. The read amplifier is used to read information from the memory disc and convert these signal pulses into logic waveforms to be entered into the computer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18 Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

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DRAWINGS

U.S. Army Munitions Command

F10526589	Assembly, Read Amplifier

PACKAGING DATA SHEET

M14002	Data Sheet, Master Packaging

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The read amplifier shall be manufactured in accordance with Drawing F10526589 and drawings pertaining thereto.

3.2 General specifications.- The read amplifier shall meet the following requirements of MIL-F-13926, and MIL-C-46357:

3.2.1 MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Workmanship

3.2.2 MIL-C-46357.

(a) Etched circuit assembly
(b) Marking
(c) Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The read amplifier shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the read amplifier for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the read amplifier shall not be damaged or adversely affected and shall meet the requirements of 3.4.

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3.3.2 Shock.- The read amplifier shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The read amplifier shall meet the requirements of 3.4.7 and 3.4.8 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The read amplifier shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The read amplifier shall meet the requirements of 3.4.7 and 3.4.8 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of +135°F.

3.3.4.2 Non-operating.- The read amplifier shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The read amplifier shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The read amplifier shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine spray of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the read amplifier shall meet the performance requirements of this specification at standard ambient temperatures between 60° and 90°F and at a barometric pressure of 28 to 32 inches of mercury with a RH up to 90%.

3.4.1 Synchronous clamping circuit (steady state).- The synchronous clamping circuit, when in the steady or static state, shall exhibit the following:

(a) The forward voltage drop across CR1 shall be less than 0.7 (volt, direct current) vdc.

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(b) The leakage current in Q2 shall be less than 15 (micro amperes) ua.

(c) The leakage current in Q7 and Q8 shall be less than 30 ua total.

(d) The base-emitter back bias on Q5, Q7, Q10, Q15, Q17 shall be at least 0.4 vdc when the transistors are in the "off" state.

(e) The base-emitter forward voltage drop in Q10 shall be less than 0.5 vdc when Q10 is forward biased.

(f) The forward voltage drop across CR4 shall be less than 0.8 vdc.

(g) The collector-emitter voltage on Q11 when it is "off" shall be 9.5 ± 2.0 vdc.

(h) The collector leakage current in Q17 shall be less than 15 ua.

(i) The saturation voltage, collector emitter, of Q2, Q7, Q10, Q11, Q15 and Q17 shall be less than 0.3 vdc.

(j) The back bias on CR9 shall be greater than 2.0 vdc when the 6-diode gate (Diodes CR5, CR6, CR7, CR8, CR9 and CR12) is "off".

(k) The back bias on diode CR7 shall be greater than 1.8 vdc when the 6-diode gate is "off".

(l) The -12 volt power supply shall be -11.7 ± 1.5 vdc.

(m) The collector-emitter voltage on Q10 and Q15 shall be 6.3 ± 1.0 vdc when they are "off".

(n) The -3 volt reference for the 6-diode gate shall be -3.0 ± 0.5 vdc.

(o) The back bias on CR5 and CR12 shall be at least 2 vdc when the 6-diode gate is "on".

(p) The forward voltage drop across the 6-diode gate shall be less than 1.7 vdc.

(q) The leakage current in CR1 and CR4 shall be less than 0.5 ua.

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3.4.2 Read amplifier and input circuit (quiescent operation).- All transistors in the read amplifier and input circuit shall be biased for linear class A operation.

3.4.2.1 Transistor Q6.- Transistor Q6 shall be biased off and the collector leakage shall be less than 15 ua.

3.4.3 Input circuit and operational amplifier a.c. voltage gain.- The input circuit and the operational amplifier shall have an a.c. voltage gain of 27.1 ± 1.8.

3.4.4 Read amplifier threshold setting.- The read amplifier shall have a threshold value of 45.0 ± 1.3 (millivolts) mv.

3.4.5 Read flip-flop (steady state).- The saturation voltages of the transistors in the "on" condition shall be less than 0.3 vdc. The transistors in the "off" condition shall have a back bias, base-to-emitter of at least 0.3 vdc and not greater than 1.5 vdc.

3.4.6 Miscellaneous diodes.

(a) The forward voltage drop across each of the following diodes shall be less than 0.6 vdc: CR25, CR26, CR28, CR30, CR31, CR32, CR33, and CR34.

(b) The leakage current in each of the following diodes shall be less than 25 ua: CR25, CR26, CR28, CR30, CR31, CR32, CR33 and CR34.

(c) The forward voltage drop across the series combination of CR19 and CR20 shall be +1.3 ± 0.1 vdc.

(d) The forward voltage crop across the series combination of CR21 and CR22 shall be +1.3 ± 0.1 vdc.

(e) The forward voltage drop across CR17 shall be 0.6 ± 0.05 vdc.

(f) The leakage current in CR18 and CR24 shall be less than 2.5 ua.

(g) The forward voltage drop across CR27 and CR29 shall be less than 0.7 vdc.

(h) The leakage current in CR27 and CR29 shall be less than 0.5 ua.

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3.4.7 Synchronous clamping circuit (dynamic operation).- An input signal to the synchronous clamping circuit shall open the 6-diode gate for 3.6 ± 0.1 u sec (microseconds). At the end of this period the 6-diode gate shall close.

3.4.7.1 Transistor Q17 shall turn "off" 0.3 ± 0.2 u sec after the 6-diode gate closes.

3.4.8 Read flip-flop and strobe switch (dynamic operation).- The strobe switch and flip-flop shall be capable of switching states at a frequency of 250 ± 25 kc (kilocycles).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I
(100 percent inspection)

	Characteristics	Requirements	Test Procedures
101.	Synchronous clamping circuit (steady state)	3.4.1	4.5.2.1
102.	Synchronous clamping circuit (dynamic state) shut-off	3.4.7 & 3.4.7.1	4.5.2.2
103.	Read amplifier and input circuit (quiescent operation)	3.4.2 & 3.4.2.1	4.5.2.3
104.	Input circuit and operational amplifier a.c. voltage gain	3.4.3	4.5.2.4
105.	Read amplifier threshold setting	3.4.4	4.5.2.5
106.	Read flip-flop (steady state)	3.4.5	4.5.2.6
107.	Read flip-flop and strobe switch (dynamic operation)	3.4.8	4.5.2.7
108.	Miscellaneous diodes	3.4.6	4.5.2.8

4.3 Control tests.

4.3.1 General.- One read amplifier shall be selected at random as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

	Control Test	Requirement	Test Procedures
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

4.3.2 Environmental.- Three units shall be selected at random as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

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TABLE III

Control Test	Requirement	Test Procedures
303. Vibration	3.3.1	4.5.1.1
304. Shock	3.3.2	4.5.1.2
305. Humidity-Operating	3.3.4.1	4.5.1.3.2
306. Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307. Salt spray	3.3.6	4.5.1.4
308. Operating temperature	3.3.3.1, 3.4.7,3.4.8	4.5.1.5,4.5.2.2, 4.5.2.7
309. Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the re-submitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

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Item	Description
1	Voltage measuring device (dc); accuracy ± 5 percent of reading, input resistance 100 K ohms or greater.
2	Voltage measuring device (ac); accuracy ± 1 percent of reading, range from 0.1 millivolts to 300 volts. Input impedance 10 M ohms and 15 pico-farads. Frequency range from 10 cps to 5 mc or greater.
3	Oscilloscope; accuracy ± 4 percent rise time 50 nanoseconds or less, sweep period from 50 nanoseconds to 1 millisecond per centimeter or greater. Input resistance 1 megohm or greater. Input capacitance 20 pico-farads or less. Dual channel preamplifier.
4	Sine wave generator; frequency 250 ± 25 kilocycles per second, output adjustable from 0 to 1 volt or greater, peak-to-peak. Distortion less than 3 percent. Output impedance less than 700 ohms.
5	Square wave generator. The fundamental frequency shall be 50KC ± 5KC and the wave shall be approximately symmetrical (each half period shall be 10.0 ± 1.0 microseconds). The two states of the square wave are 0.2 ± 0.2 volts and minus 6.5 ± 0.5 volts. The 10 to 90 percent rise and fall time of voltage transitions shall each be less than 0.4 micro-seconds. The generator output impedance shall be less than 100 ohms.
6	Two attenuating probes; attenuating factor of 10. Input resistance equal to 10 M ohms or greater. Input capacitance 10 pico-farads or less. Probe is to be used with Item 3.
7	Power supply; minus 18 volts, accuracy ± 2 percent, current rating 100 milliamps or greater, ripple equal to or less than 100 mv peak-to-peak, output impedance equal to or less than 2 ohms from D.C. to 500kc.
8	Power supply; minus 10 volts, accuracy, ± 2 percent, current rating 20 milliamperes or greater, ripple equal to or less than 100 mv peak-to-peak, output impedance equal to or less than 2 ohms from DC to 500kc.
9	Power supply; minus 6 volts, accuracy ± 2 percent, current rating 20 milliamperes or greater, ripple equal to or less than 100 mv peak-to-peak, output impedance equal to or less than 2 ohms from DC to 500kc.

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Item	Description
10	Power supply; minus 3 volts, accuracy ± 2 percent, current rating 10 milliamperes or greater, ripple equal to or less than 100 mv peak-to-peak, output impedance equal to or less than 2 ohms from DC to 500kc.
11	Power supply; minus 1.5 volts, accuracy ± 2 percent, current rating 5 ma or greater, ripple equal to or less than 2 ohms from DC to 500kc.
12	Power supply; plus 1.25 volts, accuracy ± 2 percent, current rating 20 ma, or greater, ripple equal to or less than 100 mv peak-to-peak, output impedance equal to or less than 2 ohms from DC to 500kc.
13	Power supply; plus 6 volts, accuracy ± 2 percent, current rating 20 ma or greater, ripple equal to or less than 100 mv peak-to-peak, output impedance equal to or less than 2 ohms from DC to 500kc.
14	Power supply; plus 15 volts, accuracy ± 2 percent current rating 50 ma or greater, ripple equal to or less than 100 mv peak-to-peak, output impedance equal to or less than 2 ohms from DC to 500kc.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the read amplifier rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

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4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the read amplifier shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that receives the read amplifier in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.2 and 4.5.2.7 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The read amplifier shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2.2 and 4.5.2.7 for compliance with 3.3.6.

4.5.1.5 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the read amplifier shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.2.2 and 4.5.2.7 to determine compliance with the requirements of 3.4.7 and 3.4.8 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 read amplifiers produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

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4.5.2 Performance.

4.5.2.1 Synchronous clamping circuit (steady state).- To determine compliance with 3.4.1, connect the power supplies specified in Column A to the respective pins of Column B of Table IV. Connect the ground returns of these power supplies to pins 35A and 36A. Connect pins 59B, 6B and 14B to pin 36A. Connect a 2.2K ohms ± 5 percent 1/4 watt resistor between pins 30A and 28B. There shall be no resistor in the space designated for R20.

TABLE IV

Column A Power Supply, Volts	Column B Pin
-18	54B
-10	17B and 18B
- 6	25B 26B
- 3	33B 34B
-1.5	49A 50A
+1.25	39B 40B
+6	61B 62B
+15	51A 52A

4.5.2.1.1 The following test procedures shall be conducted to determine compliance with all requirements of 3.4.1:

(a) Pin connections and voltage measurements shown in Table V shall be utilized to determine acceptable performance (Column C). Measure the voltage from the pin listed in Column A to the corresponding pin listed in Column B. The voltage shall be as indicated in Column C when energized in accordance with 4.5.2.1. Upon completion of these tests remove the 2.2K ohms resistor specified in 4.5.2.1 (between pins 30A and 28B).

TABLE V

Column A From Pin	Column B To Pin	Column C Measured Voltage vdc
14B	18A	less than +0.7
19B	17B	less than +0.01
19A	26B	less than +0.03
17A	35A	greater than +0.4
29B	35A	greater than +0.4
35A	20A	less than +0.55
20A	28B	less than +0.8
35A	44B	+9.5 ±2.0
23B	26B	less than +0.07
35B	39B	greater than +0.4

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TABLE V (Cond)

Column A	Column B	Column C	
From Pin	To Pin	Measured Voltage vdc	
35A	30B	less than	+0.3
43B	51B	greater than	+2.0
35A	22B	less than	+0.3
52B	43B	greater than	+1.8
35A	30A	+11.7	±1.5

(b) Repeat the procedures in (a) above except that Table VI shall be utilized. Upon completion of these tests remove the connection between pins 14B and 36A.

TABLE VI

Column A	Column B	Column C	
From Pin	To Pin	Measured Voltage vdc	
35A	19A	less than	+0.3
35A	30B	+6.3	±1.0
35A	44B	less than	+0.3
20A	35A	greater than	+0.4
20B	35A	greater than	+0.4
39B	22B	+6.3	±1.0
39B	23B	less than	+0.3
35A	43B	+3.0	±0.5
52B	30B	greater than	+2.0
44B	51B	greater than	+1.5
51B	52B	less than	+1.7

(c) Connect the minus 6-volt power supply through a 20K ohm ±5 percent, 1/4 watt resistor to pin 14B. Measure the voltage from pin 14B to the minus 6-volt power supply. The voltage shall be less than 0.01 vdc. Measure the voltage from pin 33B to pin 19B. The voltage shall be less than 0.3 vdc. Remove the minus 6-volt power supply and the 20K ohm resistor from pin 14.

(d) Connect the plus 6-volt power supply through a 20K ohm ±5 percent, 1/4 watt resistor to pin 28B. Measure the voltage from the plus 6-volt power supply to pin 28B. The voltage shall be less than 0.01 vdc. Remove the plus 6-volt power supply and the 20K ohm resistor from pin 28B.

4.5.2.2 Synchronous clamping circuit (dynamic operation).- To determine compliance with 3.4.7 and 3.4.7.1, the power supplies shall remain connected as specified in Table IV. The ground returns of these power supplies shall be connected to pins 35A and 36A. Connect the following pins together: 36A, 59B and 6B. Connect the

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square wave generator specified in item 5 of 4.4.1.1 to pin 14B. Connect the ground return of the generator to pin 35A. Connect one channel of the oscilloscope specified in item 3 of 4.4.1.1 using the attenuating probe specified in item 6 of 4.4.1.1, to pin 30B. Connect the ground return of the oscilloscope to pin 35A. Select resistor from Table VII, and connect in the space for R20. This resistor will be chosen so that when the square wave specified in item 5 of 4.4.1.1 is applied to pin 14B the waveform shown in Figure 1 will be observed on the oscilloscope.

TABLE VII

	Part Numbers
750 ohms	10526089
1.0 K ohms	10526090
1.24K ohms	10526091
1.5 K ohms	10526092
1.8 K ohms	10526093
2.0 K ohms	RN60C2001F
2.2 K ohms	10526094
2.32K ohms	RN60C2321F
2.37K ohms	RN60C2371F
2.43K ohms	RN60C2431F
2.49K ohms	RN60C2491F
2.55K ohms	RN60C2551F
2.61K ohms	RN60B2611F
2.7 K ohms	10526095
3.3 K ohms	10525856
3.9 K ohms	10526096
4.32K ohms	10526097
4.7 K ohms	10526098
5.1 K ohms	RN60C5101F
5.3 K ohms	RN60C5301F
5.5 K ohms	RN60C5501F
5.6 K ohms	10526099
6.2 K ohms	RN60C6201F
6.8 K ohms	10526100
7.4 K ohms	RN60C7401F
8.0 K ohms	RN60C8001F
9.1 K ohms	10526598
10.0 K ohms	10526599

4.5.2.2.1 Remove the oscilloscope probe from pin 30B and connect it to pin 44B. Connect the other channel of the oscilloscope specified in item 3 of 4.4.1.1 using an attenuating probe (item 6 of 4.4.1.1) to pin 23B. Using alternate sweeps and an external scope trigger from pin 44B, the waveforms of Figure 2 will be observed on the oscilloscope. Upon completion of this test remove the oscilloscope, square wave generator, and resistor connections from the sub-assembly.

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4.5.2.3 Read amplifier and input circuit (quiescent operation).- Compliance with the requirements of 3.4.2 and 3.4.2.1 shall be determined by the following tests with the power supplies, ground return voltages and pin connections as specified in 4.5.2.2:

(a) The voltage when measured from pin 59B to pin 65A shall be less than 0.5 vdc.

(b) With pin 65A connected to pin 43B the voltage as measured from the pins and test points (T_p) in column A to those shown in column B shall be greater or less as specified in column C of Table VIII.

Table VIII

A	B	C
Pin 65A	T_p1	less than 0.4 vdc
T_p3	T_p1	less than 0.4 vdc
T_p2	T_p3	greater than 0.2 vdc
Pin 55B	Pin 25B	less than 0.05 vdc

(c) When measuring voltage from the pin or test point in column A to pin 60B the voltage measured shall be as specified in column B of Table IX.

TABLE IX

A	B
T_p5	-2.8 ± 0.3 vdc
6	-2.6 ± 0.3 vdc
7	-2.7 ± 1.0 vdc
8	-2.4 ± 1.0 vdc
9	-5.0 ± 1.5 vdc
12	-4.7 ± 1.5 vdc
10	-7.7 ± 2.0 vdc
13	-7.4 ± 2.0 vdc
11	-11.0 ± 2.0 vdc
14	-10.7 ± 2.0 vdc
Pin 41B	+0.5 ± 0.5 vdc
50B	+0.8 ± 0.5 vdc
48B	-3.0 ± 1.0 vdc
46B	-11.0 ± 2.0 vdc
45B	-10.8 ± 2.0 vdc

(d) Remove the connection between pins 65A and 43B. Measure the voltage from pin 55B to pin 25B. The voltage measured shall be less than +6.0 ± 0.5.

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4.5.2.4 Input circuit and operational amplifier ac voltage gain.- The power supplies, ground returns and pin connections in 4.5.2.2 shall be utilized to determine compliance with 3.4.3. Connect a 22K ohm ±5 percent, 1/4 watt resistor between pins 65A and 54B. Connect the sine wave generator specified in item 4 of 4.4.1.1 through a 0.1 microfarad ±10 percent 100 volt non-polarized capacitor to pin 65A. Connect the ground return to pin 35A. Connect the oscilloscope specified in item 3 of 4.4.1.1 using the attenuating strobe (item 6 of 4.4.1.1) to T_p14. Connect the ground of the oscilloscope to pin 60B. Adjust the sine wave generator to obtain a 200 ± 60 mv, peak to peak, 250 ± 25 KC, sine wave at pin 65A. Measure this a.c. voltage from pin 65A to pin 35A with the voltage measuring device (item 2 of 4.4.1.1).

4.5.2.4.1 Measure the a.c. voltage from T_p14 to pin 35A with the voltage measuring device in item 2 of 4.4.1.1. The a.c. signal from T_p14 to pin 35A shall be from 27.1 ± 1.8 times the a.c. signal from pin 65A to pin 35A. Observe the sine wave at T_p14 on the oscilloscope. The sine wave shall be 5.5 ± 2.0 volts, peak to peak.

4.5.2.5 Read amplifier threshold setting.- To determine compliance with 3.4.4, the power supplies, pins and the generator shall remain connected as in previous test. Connect the oscilloscope specified in item 3 of 4.4.1.1 using the attenuating probe specified in item 6 of 4.4.1.1 to T_p16. Connect the ground return of the oscilloscope to pin 60B. Connect the voltage measuring device specified in item 2 of 4.4.1.1 to pin 65A. Connect the ground return to pin 60B.

4.5.2.5.1 Observe the waveform on the oscilloscope. Select resistor R58 from Table VII so that when the input sine wave is increased from zero to 45 ± 1.3 mv, zero to peak, the observed waveform is less than 0.9 volts, zero to peak. This will insure compliance with 3.4.4. When the input sine wave is increased above this value one of the waveforms shown in figure 3 will appear on the oscilloscope. Remove the sine wave generator oscilloscope, and voltage measuring device from the sub-assembly.

4.5.2.6 Read flip-flop, steady state.- To determine compliance with 3.4.5 the power supplies, ground returns and pins shall remain connected as specified in 4.5.2.2. Connect the minus 6-volt power supply to pin 11B momentarily. Measure the voltage from the pin indicated in Column A of Table X to the corresponding pin in Column B of Table X. The voltages indicated in the corresponding row of Column C must be obtained to conform with the requirement of 3.4.5.

TABLE X

A	B	C
Pin 40B	Pin 4B	less than +0.3 vdc
6B	12B	less than +0.3 vdc
40B	10B	-0.35 ± 0.2 vdc
4B	6B	+1.2 ± 0.3 vdc

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4.5.2.6.1 Connect the minus 6-volt power supply to pin 2A momentarily. Measure the voltage from the pin indicated in Column A of Table XI to the corresponding pin in Column B of Table XI. The voltages indicated in the corresponding row of Column C must be maintained to conform with the requirement of 3.4.5.

TABLE XI

A	B	C
Pin 40B	Pin 13A	less than +0.3 vdc
6B	7A	less than +0.3 vdc
5B	40B	+0.35 ± 0.2 vdc
13A	6B	+1.2 ± 0.3 vdc

4.5.2.7 Read flip-flop and strobe switch, dynamic operation.- To determine compliance with 3.4.8 the power supplies, ground returns and pins shall remain connected as specified in 4.5.2.2. Connect the sine wave generator specified in item 4 of 4.4.1.1 thru a 0.1 ± 10 percent, 100 volt, non-polarized capacitor to pin 65A. Connect the ground return to pin 35A. Adjust the output amplitude of the sine wave generator to 200 ± 60 mv peak to peak. Connect the -10V power supply to pin 1B. Connect the oscilloscope specified in item 3 of 4.4.1.1 using the attenuating probe specified in item 6 of 4.4.1.1 to pin 7A. Connect the ground return of the oscilloscope to pin 6B. The waveform shown in figure 4 will appear on the oscilloscope to insure conformance with the requirements of 3.4.9. Connect oscilloscope to pin 12B, the waveform shown in figure 4 will appear on the oscilloscope. Remove the connections from the sine wave generator and the oscilloscope to the sub-assembly. Remove the -10V power supply from pin 1B.

4.5.2.8 Miscellaneous diodes.- To determine compliance with 3.4.6 the test conditions of 4.5.2.2 (power supplies, ground returns and pin connections) shall be utilized. Connect the minus 6 volt power supply to pin 6A and then conduct voltage measurement tests in accordance with each of the following procedures to insure compliance with 3.4.6:

(a) Connect the minus 6-volt power supply to pin 2A momentarily. Measure the voltage from pin 7A to pin 1A. The voltage shall be less than 0.6V dc.

(b) Connect pin 8A to pin 6B. Measure the voltage from pin 8A to pin 8B. The voltage shall be less than 0.7 volts. Remove the connection between pins 8A and pin 6B.

(c) Connect pin 4A to pin 6B. Measure the voltage from pin 6B to pin 8B. The voltage shall be less than 0.6V dc.

(d) Connect the plus 6-volt power supply through a 2.2 K ohm ±5 percent, 1/4 watt, resistor to pin 2B. Measure the voltage from pin 2B to pin 4A. The voltage shall be

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(p) Connect pins 8A and 6B. Measure the voltage from pin 6A to pin 8B. The voltage shall be less than 0.034 vdc. Remove connections from pin 8A to pin 6B. Connect pins 5A and 6B. Measure voltage from pin 6A to pin 1A. The voltage shall be less than 0.034 vdc. Remove connection between pins 5A and 6B.

(q) Connect pins 7A and 12B to pin 6B. Measure the voltage from pin 6A to pin 8B. The voltage shall be less than +1.7 vdc. Measure the voltage from pin 6A to pin 1A. The voltage shall be less than +1.7 vdc. Remove the connection from pins 7A and 12B to pin 6B. Remove the 6 volt power supply from pin 6A.

(r) Connect the minus 6-volt power supply to pins 11B and 2A. Connect the power supply ground return to pin 6B. Connect pin 4A to pin 6B. Measure the voltage from pin 6B to pin 3B. The voltage shall be less than 0.6 vdc. Measure the voltage from pin 6B to pin 14A. The voltage shall be less than 0.6 vdc. Remove the minus 6 volts from pin 11B and 2A. Remove all external connections from the sub-assembly.

(s) Connect the minus 6-volt power supply thru a 4.7K ohm, ± 5 percent, 1/4 watt resistor to pin 4A. Connect the power supply ground return to pin 6B. Connect pins 3B and 14A to pin 6B. Measure the voltage from pin 4A to the minus 6-volt power supply. The voltage shall be less than 0.12 vdc. Remove all external connections from the sub-assembly.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- The Read Amplifier receives main memory signals from the read switches. These read signals are amplified and converted into logical signals.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of applicable level of preservation, packaging, and packing.

(c) Applicable stock number

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(d) Applicable packaging data sheet (see 5.1)

(e) A block of serial numbers to identify each Read Amplifier

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less than 0.6V dc. Remove the plus 6-volt power supply and resistor.

(e) Connect the plus 6-volt power supply through a 2.2K ohm, ±5 percent 1/4 watt, resistor to pin 3A. Measure the voltage from pin 3A to pin 4A. The voltage shall be less than 0.6V dc. Remove the plus 6-volt power supply and resistor.

(f) Connect the minus 6-volt power supply through a 2.2K ohm ±5 percent, 1/4 watt resistor to pin 2B. Measure the voltage from pin 2B to the minus 6-volt power supply. The voltage shall be less than 0.026 vdc. Remove the minus 6-volt power supply and resistor.

(g) Connect the minus 6-volt power supply through a 2.2K ohm ±5 percent 1/4 watt resistor to pin 3A. Measure the voltage from pin 3A to the minus 6-volt power supply. The voltage shall be less than 0.026 vdc. Remove the connection, from the minus 6-volt power supply to pin 3A. Remove the connection between pins 4A and 6B.

(h) Measure the voltage drop from pin 15A to pin 10A. The voltage shall be +1.3 ± 0.1 vdc.

(i) Connect the minus 6-volt power supply to pin 11B momentarily. Connect pin 5A to pin 6B. Measure the voltage from pin 5A to pin 1A. The voltage shall be less than 0.7 vdc. Remove the connection between pin 5A and pin 6B.

(j) Connect pin 4A to pin 6B. Measure the voltage from pin 6B to pin 1A. The voltage shall be less than plus 0.6 vdc.

(k) Measure the voltage from pin 10A to pin 15A. The voltage shall be +1.3 ± 0.1 vdc. Remove the connection between pin 4A and pin 6B.

(l) Measure the voltage from pin 33B to pin 15A. The voltage shall be 0.6 ± 0.05 vdc.

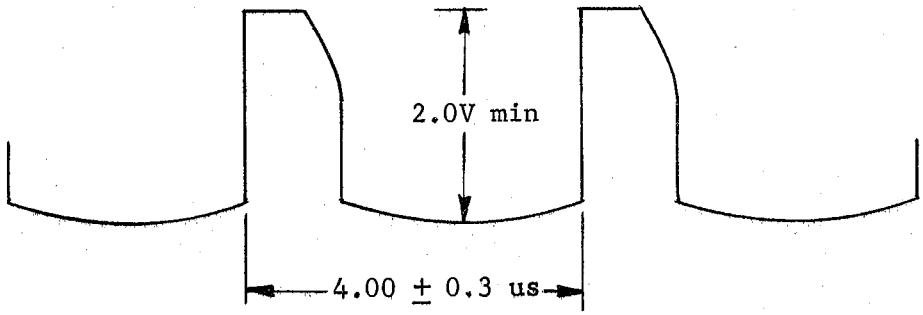
(m) Measure the voltage from pin 61B to pin 14A. The voltage shall be less than 0.3 vdc.

(n) Measure the voltage from pin 61B to pin 3B. The voltage shall be less than 0.3 vdc. Remove all external connections from the sub-assembly.

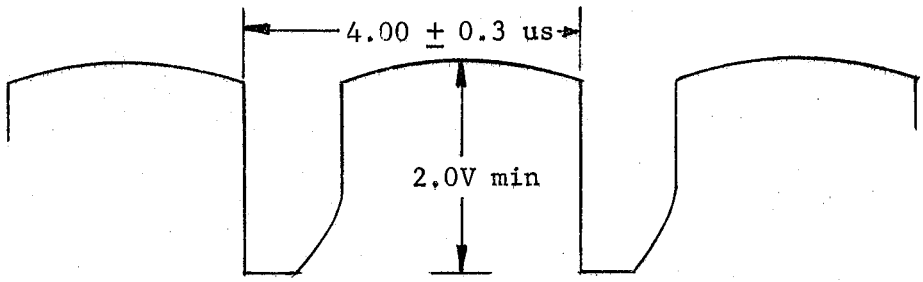
(o) Connect the plus 6-volt power supply to pin 6A. Connect the power supply ground return to pin 6B. Connect pin 4A to pin 6B. Measure the voltage from pin 6A to pin 8B. The voltage shall be less than +1.7 vdc. Remove the connection from pin 4A to pin 6B.

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a.



b.



c.

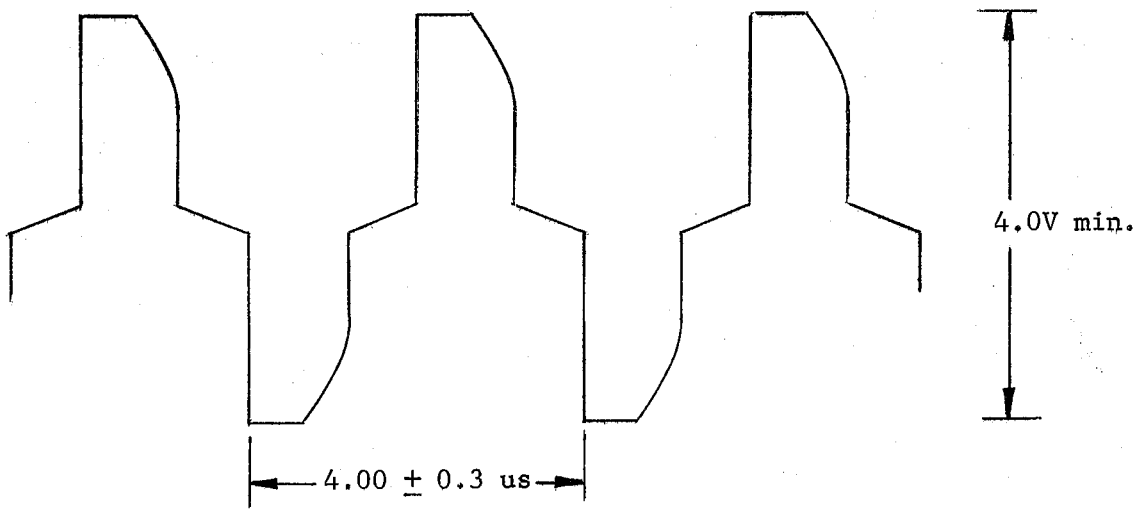


FIGURE 3

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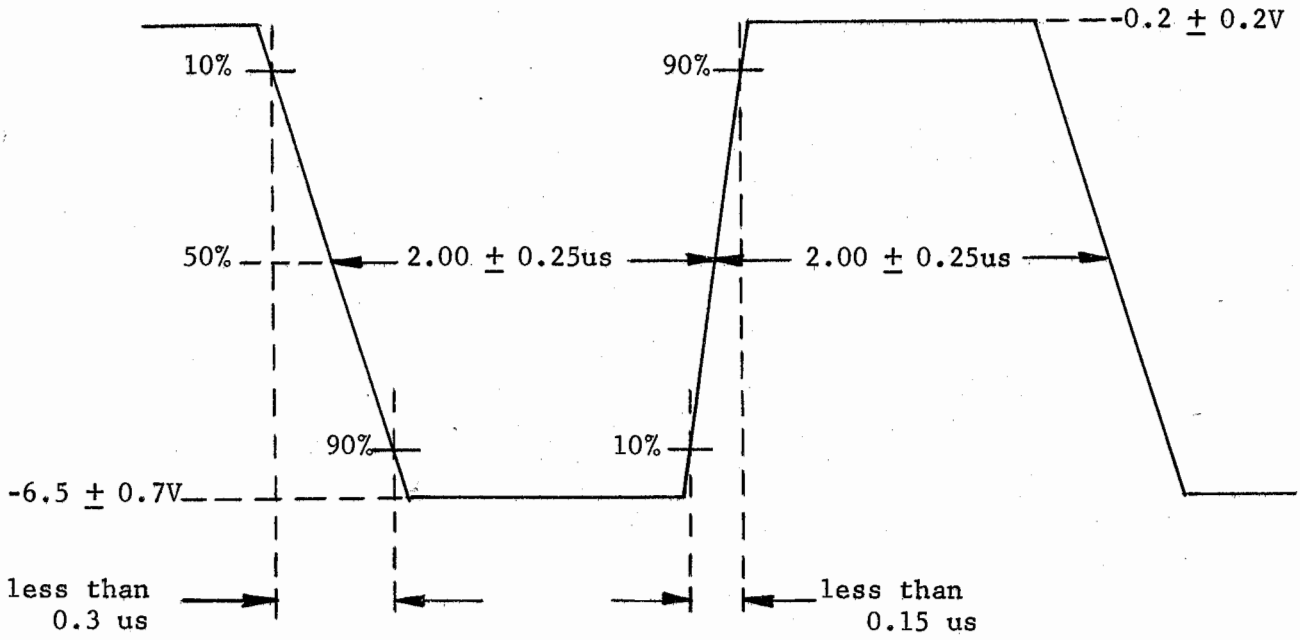


FIGURE 4

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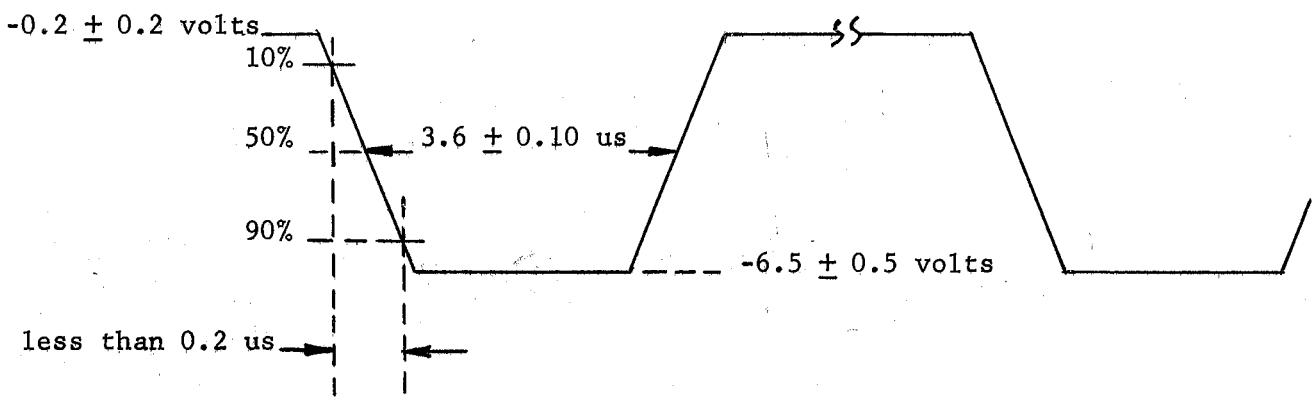


FIGURE 1

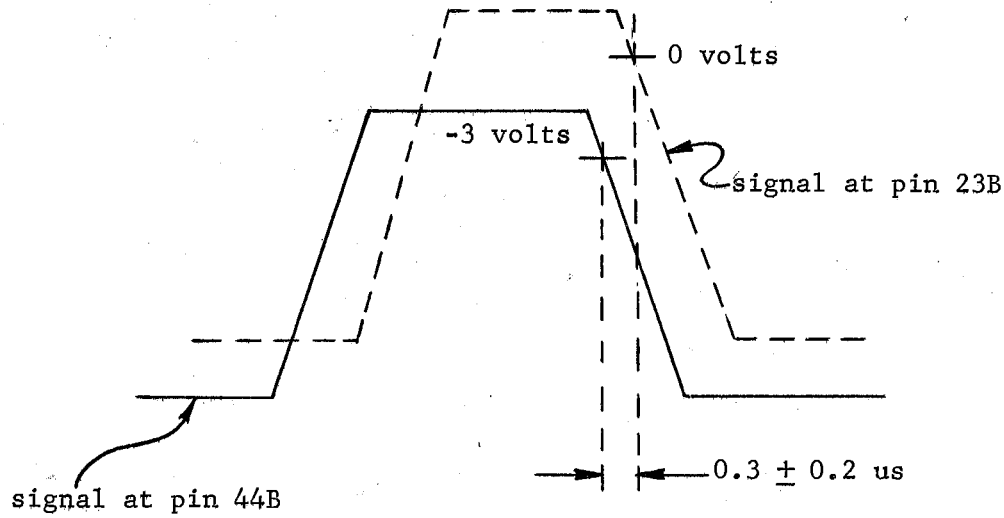


FIGURE 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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